Conditional Inductive Guardrails: Operating Boundaries of Physical Priors in Wind Forecasting

Controlled Empirical Identification Across Three Commercial Wind Farms and Electricity Market Balancing

1. Status Quo: Governance Deficit

Systematic Scoping Review (2023–2026)

N = 46 Contemporary Physics-Informed ML Studies

100.0% Uniform Positive Claims

46/46 studies report forecasting accuracy gains

The Ground Reality Ledger (Audit Findings)

- **Failure Regimes:** 0.0% reported (0/46)
- **Open Software Deficit:** 95.7% withheld (44/46)
- **Capacity Confounded:** 50.0% confounded (23/46)

Software AST Graph Divergence

Public PCT Code Reverses JS-Divergence:

Executed PyTorch: $\mathrm{JS}(y_{\mathrm{ref}}, y_{\mathrm{pred}})$ vs Formula $\mathrm{JS}(y_{\mathrm{pred}} \,\|\, y_{\mathrm{ref}})$

Induces $\mathcal{O}(1/\varepsilon)$ gradient singularity ($2.20\times$–$2.72\times$ norm surge)

The Core Research Question:

When do physical priors genuinely enhance, merely mimic, or impair operational decisions?

2. Controlled Identification Protocol

Negative Control Crucible (Capacity Strictly Matched)

Authentic Prior

Empirical Curve C_{emp}

True Aerodynamic Geometry

Pseudo-Scrambled

Disrupted Prior C_{scramble}

Form- & Weight-Matched

Information-Budget Matching Experiment

Controlled Sample Volume: $N_{\mathrm{prior}} = N_{\mathrm{train}} = 500$

- Contemporaneous sample prior retains 77% to 92% of benefit
- Proves gains stem from domain geometry, not hidden archive data

Predefined Operational Stress Axes

Penalty Sweep

$\lambda \in \{0.01, 0.1, 1.0\}$

Separates mimicry

Proxy Latency Ladder

Delay $k \in [1\mathrm{h}, 6\mathrm{h}]$

Offline staleness test

Empirical Validation Foundation (3 Commercial Wind Farms)

Kelmarsh (12.3 MW) | Hill of Towie (48.3 MW) | Penmanshiel (28.7 MW)
Coupled with 2017 UK Elexon BSC Half-Hourly Balancing Settlement

3. Operational Decision Envelope

Regime 1: Inductive Guardrail

Sample-Restricted Regime ($N_{\mathrm{train}} \leq 1{,}583$ on Penmanshiel):

- **Authentic Prior: -1.22% Downside Risk Protection**
- **Scrambled Prior: +68.1% Dispatch Penalty Surge**

Regime 2: Regularization Mimicry

Weak Penalty Schedule ($\lambda = 0.01$ on Kelmarsh):

- Authentic (0.08810) $\approx$ Scrambled (0.08801)
- Indistinguishable loss proves numerical smoothing, not physics

Regime 3: Large-Sample Inversion

High Sample Density ($N_{\mathrm{train}} \geq 5{,}000$):

- Unconstrained deep models match or outperform priors
- Fixed priors impose structural bias under rich observations

Decision-Risk Decoupling & Zero-Bid Probe

- RMSE Decoupling: Symmetric error decouples from tail risk.
- Degenerate Zero-Bid ($q \equiv 0$): Standalone cashflow earns **+£239,450/(turb·yr) on Penmanshiel without forecasting skill!**
- Proof that unhedged balancing metrics deceive market operators.

Actionable Governance: 5-Gate REFORMS-PCWPF Checklist